JNCIA-Junos Practice Lab
Project Needs & Requirements

Requirements Reference Document – Revision 1

Project location: `C:\Users\HP\jnciapracticelabs`
Date: October 28, 2026
Prepared from project owner's directives for the JNCIA-Junos CLI practice lab

1. Purpose of this document

This document stores **all the needs stated by the project owner** for completing the JNCIA-Junos Practice Lab project, together with the concrete acceptance details that those needs imply. It also pins down the **exact Junos OS look** of prompts, commands and command output that every simulated device must reproduce. A companion document (*Features Implemented*) tracks progress against these requirements.

> **Owner's directives (verbatim intent):**
> 1. Project is developed at `C:\Users\HP\jnciapracticelabs`.
> 2. "Ensure that all commands of Junos OS and devices other look like Junos OS original."
> 3. "Do make ensure that similar interface like GNS3."
> 4. "Make sure that all routers in Junos OS, commands should exactly look as Junos OS and output of all commands must also look exact as Junos OS."

2. Product vision

A browser-based **practice lab for JNCIA-Junos** certification study: a small multi-router and switch topology with fully simulated **Junos OS CLI devices** plus end hosts. The lab runs on the local machine with a simple Node.js server; **no real Junos OS images, VMs, licenses or GNS3 installation** are required. Candidates practice the JNCIA-Junos syllabus operations on an interface that looks and behaves like GNS3 with Juniper routers.

3. Master requirement register

ID	Need / requirement as stated or implied by the owner	Priority
R-01	All routers must behave like Junos OS routers; the user must be able to operate them as if the real Junos OS were running: full CLI with operational mode, configuration mode, and console login.	**MUST**
R-02	Every command that is accepted must be a real Junos OS command with real syntax.	**MUST**
R-03	The **output of every command must look exact as Junos OS** — exact prompt strings, column headers, column alignment, legends, state names, timestamps, counters and result lines.	**MUST**
R-04	CLI behavior must match Junos: Tab completion, `?` "Possible match completions" help with descriptions, exact error strings with `^` caret under the offending token, `---(more x%)---` pagination (space / b / q), pipe filters (match, except, count, last, first, no-more, display set, display xml, display json).	**MUST**
R-05	Configuration mode must reproduce real Junos workflow: candidate config model, `set/delete/deactivate/activate/rename/copy/edit/top/up/exit`, `show | compare` with `[edit]` and `+/-/!` markers, `commit`, `commit check`, `commit confirmed` with automatic rollback, `rollback 0..N`, `show system commit`, `run` , uncommitted-changes warning on `exit`.	**MUST**
R-06	Console session flow must be authentic: device boot messages, `(ttyd0)` banner, `login:`, root access to the shell (`root@R1:~ #`), and `cli` to enter the CLI.	MUST
R-07	The switching device (SW1) must present authentic Junos EX-series output (VLANs, Ethernet-switching table, RSTP bridge/interface views).	**MUST**
R-08	"Devices other" (end hosts / VPCS-class nodes) must have their own realistic console, clearly different from Junos (VPCS-style prompt `Host-A>`, VPCS-style ping/trace/show output).	MUST
R-09	The GUI must look and behave similar to GNS3: menu bar, icon toolbar, topology canvas with draggable nodes, link lines, console windows, topology summary, log, status bar, zoom and view options.	**MUST**
R-10	Consoles must open like GNS3 telnet consoles (double-click node; window titled like `R1 - telnet://127.0.0.1:5000`) and an active GNS3-like Console Manager must be available.	MUST
R-11	Node lifecycle actions like GNS3: per-node **Start / Stop / Reload** and global start-all / stop-all / reload-all; stopped nodes refuse console connections like a real telnet target.	MUST
R-12	Auto-generated per-lab topology (R1–R2–R3 routers + SW1 switch + Host-A/Host-B) must be the provided lab inventory; topology drawn on canvas with optional link labels (subnets).	MUST
R-13	Device configuration and lab layout must persist locally (config changes survive project reload; GNS3-like project save/open).	MUST
R-14	Everything must run fully offline with zero extra dependencies (Node.js only, no npm packages), startable with `START-WINDOWS.bat` / `START-MAC-LINUX.sh` / `npm start`, and remain testable with `npm test`.	MUST
R-15	Commands must produce interpretable routing effects: ping, traceroute and route tables must behave consistently with the topology, OSPF/BGP adjacency state and interface state (up/down, disabled).	MUST
R-16	Support commands like `request support information`, `request system reboot`, `start shell`, `file list/show`, cross-device `ssh/telnet` between simulated routers must exist where they exist in Junos.	SHOULD

4. Hard requirement: exact Junos OS output (R-01..R-07)

The following reference excerpts define the target strings all simulated devices must emit. These are the **canonical output formats** for the lab; spacing and legends follow real Junos OS.

4.1 Console, prompts and shell

```
Consoles: serial console. Type ^S to interrupt boot...

[Junos boot messages...]

R1  (ttyd0)

login: root
root@R1:~ # cli
root@R1>
```

```
[edit]
root@R1# show system host-name
host-name R1;

[edit]
root@R1# commit
commit complete

[edit]
root@R1#
```

4.2 Operational mode help, errors, pagination

```
root@R1> show i
Possible match completions:
  igmp-snooping       Display IGMP snooping
  ...
  interfaces          Display interface information
  ...

root@R1> show interfaces fxp
                   ^ unknown command: 'fxp'

root@R1> show interfaces fxp0 terse | match inet | count
188

---(more 47%)---
```

```
root@R1# delete system bogus
warning: statement not found
```

4.3 show interfaces terse (canonical layout)

```
Interface               Admin Link Proto    Local                 Remote
ge-0/0/0                up    up
ge-0/0/0.0              --    --   inet     10.0.12.1/30
ge-0/0/1                up    up
ge-0/0/1.0              --    --   inet     10.0.3.1/24
ge-0/0/2                down  down
fxp0                    up    up
fxp0.0                  --    --   inet     192.168.1.10/24
lo0                     up    up
lo0.0                   --    --   inet     1.1.1.1/32
```

4.4 show route (canonical layout)

```
root@R1> show route

inet.0: 10 destinations, 13 routes (10 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

0.0.0.0/0          *[Static/5] 00:23:34
                    > to 10.0.12.2 via ge-0/0/0.0
1.1.1.1/32         *[Local/8] 00:23:34, metric 0
                      Local via 1.1.1.1
2.2.2.2/32         *[OSPF/10] 00:03:12, metric 2
                    > to 10.0.12.2 via ge-0/0/0.0
10.0.12.0/30       *[Direct/0] 00:23:34, metric 0, tag 0
                    > via ge-0/0/0.0
10.0.12.1/32       *[Local/8] 00:23:34
                     Local via 10.0.12.1
...
172.16.9.0/24      *[Static/5] 00:23:34
                      Discard
```

4.5 OSPF / BGP canonical views

```
root@R1> show ospf neighbor
Address          State             Interface     Dead time  ID               Pri Retx
10.0.12.2        Full              ge-0/0/0.0    00:00:35   2.2.2.2            128
```

```
root@R1> show ospf interface
Interface           State     ID            Cost    DR          BDR         Type    MTU
ge-0/0/0.0          BDR       10.0.12.1     10      10.0.12.2   10.0.12.1   P2P     1500
```

```
root@R1> show bgp summary
Groups: 1 Peers: 1 Down peers: 0
Unconfigured peers: 0
Table          Tot Paths  Act Paths Suppressed  History Damp State    Pending
inet.0                    0          0          0        0          0        0
Peer                     AS      InPkt     OutPkt    OutQ   Flaps Last Up/Dwn
State|#Active/Received/Accepted/Damped
2.2.2.2               65000       68        68        0      0       53 Establ
```

4.6 ARP / LLDP / chassis / version canonical views

```
root@R1> show arp
MAC Address       Address         Name                      Interface     Flags
02:1c:99:xx:xx:xx 10.0.12.2       10.0.12.2                 ge-0/0/0.0    none
```

```
root@R1> show version
Hostname: R1
Model: mx240
Junos: 20.4R2-S2.1
JUNOS OS Kernel 64-bit  [20.4R2-S2.1]
JUNOS OS Crypto 60-bit  [20.4R2-S2.1]
JUNOS base OS boot [20.4R2-S2.1]
JUNOS Routing Software [20.4R2-S2.1]...
```

```
root@R1> show system uptime
Current time:      2026-10-28 06:03:12 UTC
System booted:     2026-10-28 05:10:00 UTC (00:53:12 ago)
Protocols started: 2026-10-28 05:10:55 UTC (00:52:17 ago)
Last configured:   2026-10-28 05:58:40 UTC (00:04:32 ago) by root
 5:03am  up 53 mins, 1 user, load averages: 0.07, 0.02, 0.01
```

4.7 Configuration rendering (braces / set / compare)

```
root@R1# show
version 20.4R2-S2.1;
system {
    host-name R1;
    time-zone UTC;
}
interfaces {
    ge-0/0/0 {
        description "LINK TO R2";
        unit 0 {
            family inet {
                address 10.0.12.1/30;
            }
        }
    }
}...
```

```
root@R1# show interfaces ge-0/0/0 | compare
[edit]
-   description "LINK TO R2";
+   description "LINK TO R2-NEW";
```

4.8 ping / traceroute canonical output

```
root@R1> ping 10.0.12.2 count 3
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=63 time=2.224 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=63 time=1.987 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=63 time=2.056 ms

--- 10.0.12.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max/stddev = 1.987/2.089/2.224/0.098 ms
```

```
root@R1> traceroute 3.3.3.3
traceroute to 3.3.3.3 (3.3.3.3), 30 hops max, 40 byte packets
 1  10.0.12.2 (10.0.12.2)  3.234 ms  2.567 ms  2.987 ms
 2  3.3.3.3 (3.3.3.3)  5.432 ms  5.123 ms  6.012 ms
```

5. GNS3-like interface requirements (R-09..R-12)

5.1 Shell (window chrome)

- **Menu bar:** File, Edit, View, Simulation, Tools, Help.
  File = New project / Save / Reset lab; View = zoom helpers, show grid, show link labels, topology summary; Simulation = Start/Stop/Reload all nodes; Help = About / shortcuts.
- **Toolbar** in GNS3 style: new/open/save project, snapshot, drawing tools (rectangle, ellipse, line, text note), fit view, zoom in/out/reset, grid toggle, link labels toggle, console connector, quit + start-all / stop-all / reload-all node buttons.
- **Status bar**: "Ready", zoom percent, node count, project name.

5.2 Topology canvas

- Dark GNS3-like canvas; draggable **router** icons (R1, R2, R3), **switch** icon (SW1) and **VPCS PC** icons (Host-A, Host-B), each with a device label beneath, and a colored status dot (running / stopped) derived from node state.
- Straight link lines between node icons; optional reversible **link/subnet labels** (10.0.12.0/30, 10.0.23.0/30, VLAN 10 trunk link).
- Zoom & pan; grid overlay toggle; fit-to-view button.
- Node **context menu** (GNS3-like items): Start, Stop, Reload, Console (opens a console window tied to the node).
- **Double-click node opens its console** (authentic GNS3 behavior).

5.3 Console windows / console manager

- Floating, draggable, resizable black terminal windows titled like `R1 - telnet://127.0.0.1:5000`; window title includes device name and telnet-style port.
- **Console Manager** panel tracks all open consoles with ports, and can open the consoles for devices not yet connected.
- Console of a **stopped** node shows telnet refusal (*can't open a connection...*).
- Terminal behavior: typed echo on the prompt line, **tab completion**, `?` completions listing, command history (up/down), key paging via space / b / q on `---(more)---`, Ctrl+C to abort, config-mode prompt includes the `[edit ...]` block header.

5.4 Right dock: Topology Summary + Servers Summary

- Topology summary lists each node with expandable link entries like `Link 1: ge-0/0/0 (R2 ge-0/0/0)`.
- Servers summary lists the lab server (`local: http://localhost:3000`) and its running state.

6. CLI command scope (JNCIA-Junos syllabus surface)

6.1 Operational mode

Required command families (all output exact per Section 4):

- **Diagnostics:** `ping`, `traceroute`, `monitor interface`;
- **Inventory/state:** `show interfaces` (terse / brief / media / detail / statistics), `show lldp neighbors`, `show arp`, `show vlans`, `show ethernet-switching table`, `show spanning-tree bridge/interface`;
- **Routing:** `show route` (+ exact / detail / extensive / protocol), `show route summary`, `show route forwarding-table`;
- **Protocols:** `show ospf neighbor/interface/database/route/overview/statistics`, `show bgp summary/neighbor`;
- **System:** `show version`, `show system uptime/commit`, `show chassis hardware/routing-engine/fpc pic-status/alarms/environment`, `show log messages`, `show configuration`;
- **Administration:** `configure`, `request support information`, `request system reboot`, `start shell`, `file list|show`, cross-device `ssh` / `telnet`, `clear` subcommands, `set cli screen-length`, `exit|quit`.

6.2 Configuration mode

Candidate model with `/` semantics, levels via `edit/top/up/exit`, statements `set / delete / deactivate / activate / rename / copy / annotate`, displays `show`, `show | compare [rollback n]`, `show | display set|xml|json`, commits `commit / commit check / commit confirmed / commit and-quit / commit comment "..."`, `rollback [n]`, `show system commit`, `run`, `status`.

7. Lab inventory that must exist (R-12)

Device	Type / model	Junos version	Key interfaces / addressing
R1	MX240 router	20.4R2-S2.1	ge-0/0/0 10.0.12.1/30 (R2), ge-0/0/1 10.0.3.1/24 (SW1), lo0 1.1.1.1/32, fxp0 192.168.1.10/24, OSPF area 0, static default, BGP to 2.2.2.2
R2	MX240 router	20.4R2-S2.1	ge-0/0/0 10.0.12.2/30 (R1), ge-0/0/1 10.0.23.1/30 (R3), lo0 2.2.2.2/32, OSPF area 0, BGP to 1.1.1.1
R3	MX240 router	20.4R2-S2.1	ge-0/0/0 10.0.23.2/30 (R2), lo0 3.3.3.3/32, OSPF area 0, static discard 172.16.9.0/24
SW1	EX4300-24T switch	20.4R2-S2.1	ge-0/0/0 uplink to R1 (VLAN staff trunk), ge-0/0/1 Host-A, ge-0/0/2 Host-B, irb.10 10.0.3.2/24, irb.20 10.1.1.2/24, VLANs staff(10)/guest(20), RSTP
Host-A	VPCS-style PC	–	10.0.3.20 on VLAN staff via SW1 ge-0/0/1
Host-B	VPCS-style PC	–	10.1.1.20 on VLAN guest via SW1 ge-0/0/2

8. Devices other than Junos (R-08)

End hosts must behave like **VPCS consoles**, not like Junos: prompt `Host-A>`, VPCS-style `ping` / `trace` output, `show` / `show ip` / `show arp`, `clear`, gateway behavior, and `quit`. The lab engine must still carry their traffic consistently with the Junos side (ping Host-A from R1 works over the bridged VLAN; ping across VLANs is routed by SW1's irb interfaces).

9. Delivery, quality and acceptance

- Fully offline; runs with plain Node.js (≥14); zero npm dependencies; no admin rights.
- `npm start` / launcher scripts serve the lab at `http://localhost:3000`.
- `npm test` executes automated checks over the same engine modules used by the browser UI.
- Acceptance criteria audit (must all be true before release):

No.	Acceptance criterion
A1	Every prompt string matches Junos (login/banner/shell/op/config).
A2	Every `show *` output column layout, legend and value spelling matches Section 4 samples.
A3	`? / Tab` completions list real Junos command names with descriptions.
A4	Error strings are the real Junos strings (caret positioning included).
A5	Pager `---(more x%)---` supports space / b / q exactly.
A6	Config candidate/commit/rollback behavior matches Junos semantics incl. `commit confirmed` auto-rollback.
A7	GNS3-like UI: canvas, dragging, context menu, consoles with telnet-style titles, topology summary, status bar present.
A8	VPCS hosts produce VPCS-style (not Junos) consoles.
A9	Configs + layout persist across reloads; Reset lab returns the baseline.
A10	`npm test` passes with green results covering the engine and CLI machine.